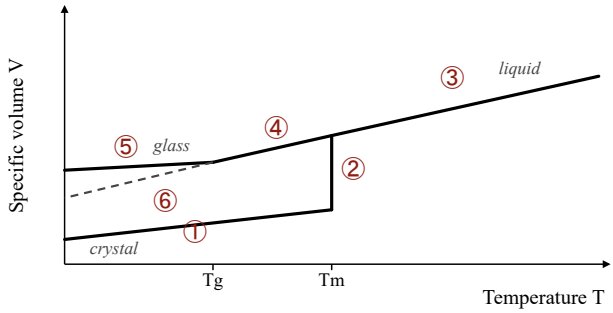
Advanced Materials — Exam Answer Sheet

42 past-exam questions · concise English answers · ①②③ in the text match the numbers on each sketch (solid = draw this; dashed = reference).

1. Thermal transitions

Q21 *What's the definition of state function? Give the physical meaning of the second-order transition in terms of ΔV and ΔH?*

State function: a property that depends only on the present state, not on the path (U, H, S, G, V; not Q or W).



① Crystal: V rises slowly with T.

② At Tm, V and H **jump** → **first-order** transition (ΔV ≠ 0, ΔH ≠ 0, latent heat).

③ Liquid: larger slope (free volume grows).

④ Fast cooling: the liquid line continues below Tm (supercooled liquid).

⑤ At Tg the line only **bends**: **ΔV = 0, ΔH = 0** (chains are frozen in place, no rearrangement), but the slopes α = (1/V)(∂V/∂T) and Cp = (∂H/∂T) change → **second-order** transition.

⑥ (dashed) the liquid line if it were not frozen.

Q9 *How to determine the Tg of an amorphous polymer (give at least 2 ways)? Why a peak appears at the Tg point as the DMA spectrum was shown?*

Ways to determine Tg:

① Dilatometry (V–T): the slope of V vs T changes at Tg.

② DSC: a step (ΔCp) in the heat-capacity baseline at Tg.

③ DMA: G′ drops ~10³-fold from glassy to rubbery; tan δ peaks at Tg.

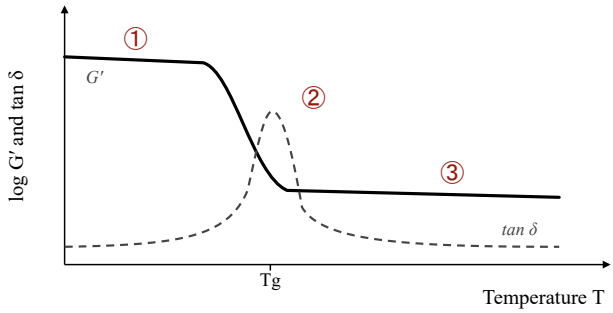
Why a DMA peak: in the glass-transition region cooperative segmental motion has just started but is sluggish, so it lags the oscillation and dissipates the most energy; G″ peaks while G′ falls, so tan δ = G″/G′ shows a maximum (τ ≈ 1/ω). See the sketch in Q37.

Q37 *Why a peak appears at the Tg point as the DMA spectrum was shown?*

① Below Tg (glassy): segments are frozen → G′ high, little energy loss → tan δ small.

② At Tg: cooperative segmental motion starts but is sluggish (τ ≈ 1/ω) → segments move yet lag the load → maximum energy dissipated as heat (G″ peaks) while G′ drops ~10³-fold → **tan δ = G″/G′ peaks**.

③ Above Tg (rubbery): segments follow the load easily → loss small again → tan δ falls.



Q19 *The melting temperature of PVDF is 174 °C, please estimate its T_g value.*

Tg ≈ 224 K (≈ −49 °C).

Empirical rule: Tg ≈ ½–⅔ Tm (in K). Tm = 174 + 273 = 447 K. PVDF is symmetric → use ½: Tg ≈ 447/2 ≈ 224 K (tabulated ≈ −45 °C).

Q8 *Please use Simha-Boyer equation to estimate the thermal expansion coefficient α_g of PS below Tg (Tg = 373 K, α_l = 5.5×10^{-4} K^{-1})*

αg ≈ 2.47×10⁻⁴ K⁻¹.

Simha–Boyer: $(\alpha l - \alpha g)Tg = 0.113$

$\alpha g = \alpha l - 0.113/Tg = 5.5\times10^{-4} - 0.113/373 = 5.5\times10^{-4} - 3.03\times10^{-4} = 2.47\times10^{-4}$ K^{-1}

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2. Structure and Tg

Q4 *Which one of the following polymers has the highest T_g (1) PS (2) Poly(vinyl biphenyl) (3) Poly(vinyl naphthalene), explain.*

(2) Poly(vinyl biphenyl), Tg ≈ 145 °C.

The bulkier the pendant group, the greater the steric hindrance to bond rotation, the higher Tg: PS (phenyl) 100 °C < PVN (naphthyl) 135 °C < PVBP (biphenyl) 145 °C.

Q5 *Which one of the following polymers has the highest T_g (1) poly(methyl acrylate) (2) poly(methyl methacrylate) (3) poly(pentyl acrylate)*

(2) PMMA (Tg ≈ 378 K).

PMMA vs PMA: the extra α-CH_3 hinders backbone rotation → PMMA 378 K > PMA 279 K.

Poly(pentyl acrylate): long flexible side chain → more free volume → lowest Tg.

Q6 *Which one of the following polymers has the highest T_g (1) polyethylene (2) polypropylene (3) polybutylene (4) polyhexylene (5) poly(4-methyl pentene)*

(5) Poly(4-methyl pentene), Tg ≈ 302 K.

Its branched, bulky side group hinders bond rotation. Linear alkyl side chains are flexible and add free volume, so Tg falls as they lengthen (PP 253 > PB 249 > polyhexene 223 K); PE (no side group) 188 K.

Q7 *Which one of the following polymers has the lowest T_g (1) PMMA (2) PEMA (3) PPMA (4) PBMA*

(4) PBMA (Tg ≈ −2 °C).

Each extra CH_2 in the ester side chain lowers Tg: PMMA 105 > PEMA 65 > PPMA 38 > PBMA −2 °C. Longer flexible side chains loosen chain packing and increase free volume (internal plasticization).

Q25 *Which one of the following PU (with the same MW and diol content) should have the lowest Tg (1) HMI (2) H_{12}MDI (3) MDI (4) TDI*

(1) HMI has the lowest Tg (HMI < H_{12}MDI < MDI ≈ TDI).

HMI is aliphatic –$(CH_2)_6$– with no ring → most flexible chain. Cycloaliphatic (H_{12}MDI) and aromatic (MDI, TDI) rings stiffen the hard segment → higher Tg.

Q10 *Estimate the Tg of 30 wt% di-n-butylphthalate (DBP) (Tg = 178 K) compounded with 70 wt% PVC (Tg = 355 K)?*

Tg ≈ 273 K (0 °C).

Fox equation: $1/Tg = w_1/Tg_1 + w_2/Tg_2 = 0.30/178 + 0.70/355 = 3.657\times10^{-3}$ K^{-1} → Tg ≈ 273 K. The plasticizer lowers Tg of PVC from 355 K.

Q11 *Which ones of the followings are miscible blends (multiple choices) (1) PC/PBT (2) PS/PPO (3) PVDF/PMA (4) PEO/PMMA*

(2), (3), (4) are miscible.

PS/PPO, PVDF/PMA and PEO/PMMA form one phase → a single Tg. (1) PC/PBT is immiscible → phase separated, two Tgs. Mixing needs $\Delta G_m = \Delta H_m - T\Delta S_m < 0$; ΔS_m of polymers is tiny, so specific interactions ($\Delta H_m < 0$) are required.

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3. Structure and Tm

Q16 *The PTFE has lower melt energy than the PE does. (2860 J/mole of PTFE vs. 3870 J/mole of PE) How come the Tm of PTFE is much higher than PE? (645 K vs. 410 K) (Hint: Table 11.1 can help you)*

Because PTFE has a much smaller melting entropy ΔSf.

$T_m = \Delta H_f/\Delta S_f \rightarrow \Delta S_f = \Delta H_f/T_m$: PE 3870/410 = 9.44; PTFE 2860/645 = 4.43 J mol^{-1} K^{-1} (about half).

PTFE is a stiff helical chain (bulky F atoms, large trans–gauche energy gap) → few conformations gained on melting → small ΔSf → high Tm despite lower ΔHf.

Q12 *Why is the Tm of isotactic PP higher than that of syndiotactic PP (187 vs. 150)?*

Tm = ΔHf/ΔSf. Both iPP and sPP are regular and crystallize, but:

iPP packs its 3_1 helices more tightly → larger ΔHf.

sPP forms thinner, less perfect lamellae → Thompson–Gibbs $T_m = T_m^\circ[1 - 2\sigma_e/(l\cdot\Delta h_f)]$: smaller l → lower Tm.

So iPP 187 °C > sPP 150 °C.

Q22 *Will the "regularity of configuration" affect more on Tg or Tm? Please use PP as an example to illustrate your point.*

Regularity (tacticity) affects Tm much more than Tg.

Tm: iPP and sPP are regular → crystallize → have Tm; atactic PP cannot crystallize → no Tm.

Tg: depends on chain stiffness / pendant-group size; all three PP have the same –CH_3 → Tg ≈ −10 °C for all.

Q13 *Place the following polyurethane structures in order of decreasing Tm? (1) –HN–$(CH_2)_6$–NH–CO–O–$(CH_2)_3$–C_6H_4–$(CH_2)_3$–O–CO– (2) –HN–$(CH_2)_8$–NH–CO–O–$(CH_2)_2$–C_6H_4–$(CH_2)_2$–O–CO– (3) –HN–$(CH_2)_8$–NH–CO–O–$(CH_2)_{10}$–O–CO–*

(1) > (2) > (3) (≈ 513 > 481–485 > 406 K), by $T_m = \Delta H_f/\Delta S_f$.

① (1), (2): phenylene ring stiffens the chain → small ΔSf → high Tm.

② (3): no ring, 18 flexible CH_2 → large ΔSf; longest repeat unit → fewest urethane H-bonds → small ΔHf → lowest.

③ (1) vs (2): same ring and CH_2 count; the shorter $(CH_2)_6$ diamine in (1) concentrates H-bonding → larger ΔHf → (1) > (2).

Q14 *Place the following Tm of the polyolefins with different side chain structures in a decreasing order and give a brief explanation (1) –CH_3 (2) –CH_2–$C(CH_3)_2$–CH_2CH_3 (3) –$CH_2CH_2CH_3$ (4) –$CH_2CH_2CH_2CH_3$ (5) –CH_2–$CH(CH_3)$–CH_2CH_3*

(2) > (5) > (1) > (3) > (4), by $T_m = \Delta H_f/\Delta S_f$.

Bulky branched side groups near the backbone hinder rotation → stiff chain → small ΔSf → high Tm; (2) (quaternary C) is more crowded than (5).

Linear flexible side chains: longer → larger ΔSf → lower Tm (CH_3 > propyl > butyl).

Q15 *Please list the Tm of the followings with the same MW in a decreasing order: (a) –[Ph–CO–Ph–O]– (b) –[Ph–CO–Ph–CO–Ph–O]– (c) –[Ph–O–Ph–CO–Ph–O]– (d) –[Ph–Ph–O–Ph–CO–Ph–O]–*

(d) > (b) > (a) > (c) (689 > 657 > 640 > 628 K).

Stiffer backbone → smaller ΔSf → higher Tm = ΔHf/ΔSf. (d) has a directly linked biphenyl (stiffest); (b) has one more rigid C=O than (a); (c) has two flexible ether –O– links → lowest.

Q17 *Place the following Tm of nylon in a decreasing order and give brief explanation, Nylon 8, Nylon 9, Nylon 10, and Nylon 11.*

Nylon 9 > 8 > 11 > 10 (482 > 473 > 463 > 461 K).

Larger n → lower amide density → fewer interchain H-bonds → lower ΔHf → lower Tm. Odd–even effect: odd-n nylon 9 aligns its H-bonds better, so 9 > 8; 11 and 10 differ by only ~2 K.

Q18 *Place the following Tm of polyester in a decreasing order $[(CH_2)_mCOO]$; m = 2, 3, 4, 5, 6*

m = 2 > 3 = 5 > 4 > 6 (395 > 335 = 335 > 329 > 325 K).

Larger m → lower ester density and more flexible CH_2 → ΔHf↓, ΔSf↑ → Tm falls; m = 4 dips by the odd–even effect.

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4. Crystallinity and barrier properties

Q3 *How to determine the crystallinity of a polymer?*

① Density: φc = (ρ − ρa)/(ρc − ρa) (crystals are denser).
② X-ray diffraction (WAXD): area of sharp crystalline peaks vs the amorphous halo.
③ DSC: Xc = ΔHm/ΔHm° (melting-endotherm area ÷ 100 %-crystal value).

Q20 *How to determine the crystallinity of a polymer? How to determine the lamellae thickness of a polymeric crystal?*

Crystallinity: density φc = (ρ − ρa)/(ρc − ρa); WAXD (crystalline peaks vs amorphous halo); DSC (ΔHm/ΔHm°).

Lamellar thickness l:

① DSC Tm + Thompson–Gibbs: Tm = Tm°[1 − 2σe/(l·Δhf)] → l = 2σeTm°/[Δhf(Tm° − Tm)]; thinner lamellae melt lower.
② SAXS: long period L = λ/(2 sin θ), l ≈ φc·L.
③ TEM / AFM imaging; Raman LAM. Typical l ≈ 10–20 nm.

Q38 *Which of the following polymers would expect to have the best barrier properties (hint: crystalline materials perform better than non-crystalline materials) (1) Atactic polystyrene (2) A random ethylene/propylene copolymer (50/50) composition. (3) mCOC (4) HDPE*

(4) HDPE. Crystallization requires a regular chain.

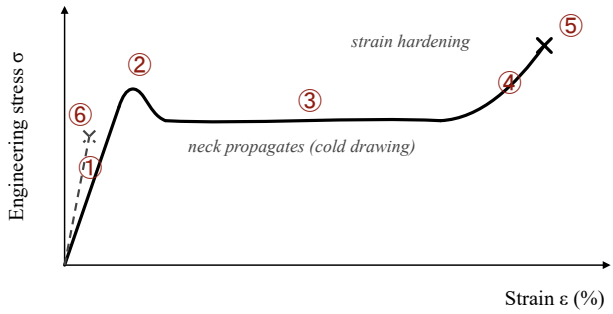
① HDPE: linear $–CH_2–$ chain, no side groups, few branches → packs easily → highest crystallinity (ρ = 0.94–0.97 g cm^{-3}).
② Atactic PS: randomly placed phenyl groups → amorphous.
③ Random E/P copolymer: random sequence → amorphous.
④ mCOC: bulky random cyclic units → amorphous.
⑤ Crystallites are impermeable (lower S) and make a tortuous path (lower D) → P = D·S lowest for HDPE.

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5. Stress–strain curves

Q1 *Please give a whole picture of the stress-strain curves for a semi-crystalline polymer with a polymer-morphology point of view.*

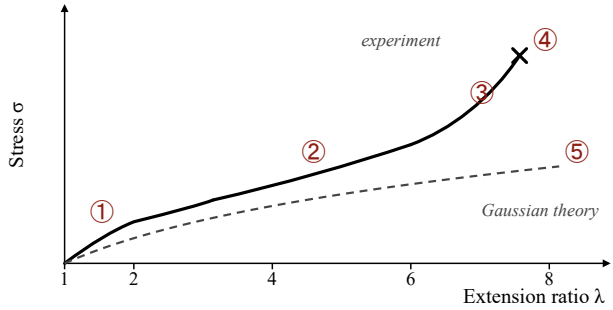
① Linear, steep: amorphous tie chains stretch, lamellae bend slightly (elastic, $\sigma = E\varepsilon$).
② Yield point, then stress drops: lamellae tilt/slip, a neck forms.
③ Plateau: the neck propagates at nearly constant stress (cold drawing); lamellae break into blocks and align into microfibrils.
④ Strain hardening: chains are fully oriented, load carried by backbones.
⑤ Fracture (×).
⑥ (dashed) below Tg or fast pulling: brittle fracture, no yield.



Q2 *Please give a whole picture of the stress-strain curves for a rubbery polymer with a molecular conformation point of view.*

Entropic elasticity: stretching reduces conformational entropy → retractive force, $f = NkT(\lambda - \lambda^{-2})$.

① Low λ: obeys the Gaussian theory, low modulus — coiled chains are just uncoiled.
② $\lambda \gtrsim 3$: stress rises above theory (Mooney–Rivlin fits better).
③ Large λ: steep rise — chains are nearly straight but still crosslinked (strain-induced crystallization in NR).
④ Fracture at $\lambda \approx 7$–8. Fully reversible up to break; no yield, no necking.
⑤ (dashed) Gaussian theory line.



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6. Rubber elasticity

Q39 *In deriving the rubber elastic theory, what assumptions were used.*

① Gaussian chains (long chains, end-to-end distance Gaussian).
② Force is purely entropic; internal-energy change ignored ($\Delta U = 0$).
③ Affine deformation (crosslinks move with the macroscopic strain).
④ No volume change: $\lambda_1\lambda_2\lambda_3 = 1$.
⑤ Ideal network: no dangling ends or loops.
⑥ Isothermal and reversible.

Result: $f = NkT(\lambda - \lambda^{-2})$.

Q33 *Pick up the right choices as follows (multiple choices) (1) Endothermic under tension (2) Modulus increases with temperature (3) Elasticity comes mainly from the entropy change (4) Adiabatic process is suitable to describe rubber under a large deformation?*

(2), (3), (4).

(1) Wrong: stretching lowers entropy → heat is released (exothermic).

(2) Right: $f \approx -T(\partial S/\partial l) \rightarrow$ modulus rises with T.

(3) Right: elasticity comes from the entropy decrease.

(4) Right: fast, large extension is reversible adiabatic (no time for heat exchange) → rubber warms up.

Q34 *Pick up the right choices as follows (multiple choices) for $f = (\partial U/\partial l)_T - T(\partial S/\partial l)_T$ (1) $(\partial U/\partial l)_T$ is valid for crystalline and glassy state (2) $(\partial U/\partial l)_T$ represents the Hook's elasticity (3) $T(\partial S/\partial l)_T$ represents the Rubber elasticity (4) $(\partial U/\partial l)_T$ increases with temperature (5) For a given elongation, the $(\partial U/\partial l)_T$ is 10^4 higher than the $T(\partial S/\partial l)_T$.*

(1), (2), (3).

(1)(2) Right: crystals and glasses deform by bond stretching, $f \approx (\partial U/\partial l)_T$ = Hookean (energy) elasticity.

(3) Right: $-T(\partial S/\partial l)_T$ is rubber (entropy) elasticity.

(4) Wrong: the energy term is nearly independent of T; only the entropy term grows with T.

(5) Wrong: in rubber the energy term is the smaller one ($f_U/f \approx 0.1$–0.2).

Q42 *Which one of the following statement regarding polyisoprene (natural rubber) is right (1) Natural rubber is mainly a trans form (2) The trans form has lower Tg (3) the repeat distance of cis is 0.88 nm (4) The Golfball covers can be made by Polyisoprene in trans form*

Only (4).

(1) Wrong: natural rubber is cis-1,4-polyisoprene.

(2) Wrong: trans has the higher Tg (220 vs 200 K) and Tm.

(3) Wrong: cis repeat distance is 0.81 nm; 0.88 nm is trans.

(4) Right: hard, crystallizable trans-polyisoprene (gutta-percha) is used for golf-ball covers.

Q35 *Please give your brief reasons why the space shuttle Challenge exploded in 1988.*

The launch (1986) was on a very cold morning. At low temperature the fluoroelastomer O-ring became stiff and lost its resiliency: it recovered its shape too slowly (Feynman's ice-water demonstration). When the joint opened, the O-ring could not reseal → hot gas leaked and eroded the seal → the flame ignited the hydrogen fuel in the external tank.

Q36 *What is the "manufacturing secret" to make the "thermoplastic elastomer"?*

A triblock copolymer: hard–soft–hard (e.g. SBS: PS–PB–PS).

① Hard end blocks (glassy or crystalline) phase-separate into domains = **physical crosslinks**.

② The low-Tg soft middle block links the domains → rubber elasticity at use temperature.

③ Above the hard-block Tg (or Tm) the domains soften → it flows and can be processed like a thermoplastic; the network re-forms on cooling (recyclable).

Q24 *Why does a SBS triblock copolymer exhibit a non-Newtonian and even highly elastic behavior at low temperature (< 423 K) but show a Newtonian behavior at 448 K?*

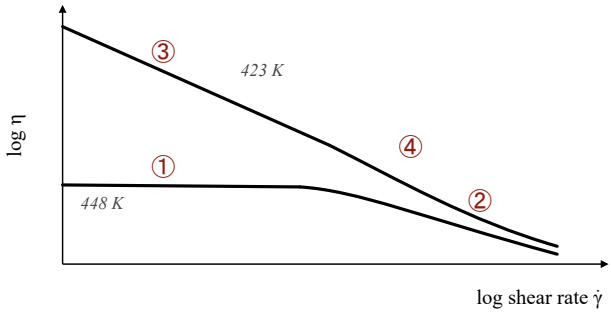
Key: PS domains act as physical crosslinks at low T and disappear at high T.

① 448 K (above the order–disorder transition): PS and PB mix, no network → Newtonian plateau η_0 at low $\dot{\gamma}$.

② 448 K: thins only at high $\dot{\gamma}$, like a normal melt.

③ 423 K: PS blocks stay segregated in domains → physical network → yield-stress-like behaviour, no plateau ($\eta \approx \tau y/\dot{\gamma}$, slope ≈ −1): non-Newtonian and elastic.

④ 423 K, high $\dot{\gamma}$: the network is broken → η drops toward the 448 K curve.



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7. Viscoelasticity

Q27 *List four elastic characters of a viscoelastic polymer.*

① Rod climbing (Weissenberg effect): the melt climbs a rotating rod (normal stresses); a Newtonian fluid dips.

② Die swell: the extrudate is thicker than the die.

③ Tubeless siphon: flow continues after the tube is lifted out of the liquid.

④ Secondary flow / vortices in a contraction (also: elastic recoil).

Cause: stretched chains store elastic energy (fading memory); Newtonian fluids show none of these.

Q31 *Pick up the right choices for describing the Relaxation Modulus vs. Temp. (Master curve) as follows (multiple choices) (1) Glassy modulus is 10^2 to 10^3 times higher than the Rubbery Modulus (2) Rubbery Modulus increases with the crosslinking degree (3) Rubbery Modulus increases with the crystallinity (4) the modulus of semicrystalline polymer remains independent of temperature if the crystallinity remains unchanged. (5) The crystallinity has little effect on the Glassy modulus*

All five, (1)–(5), are correct.

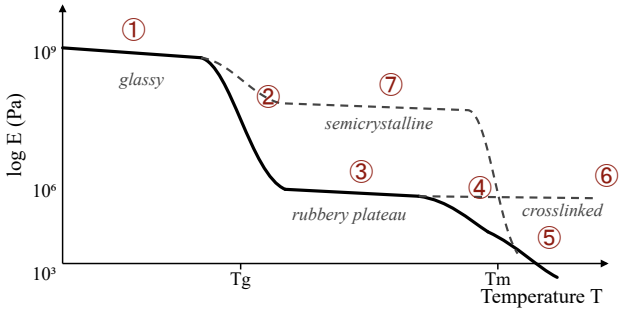
(1) Glassy ~10^9 Pa (①) vs rubbery ~10^6 Pa (③): 10^2–10^3 times.

(2) Crosslinks keep the plateau (⑥, no flow); more crosslinks → higher plateau ($G = \rho RT/Mc$).

(3) Crystallites act as filler and crosslinks → higher modulus between Tg and Tm (⑦).

(4) Constant crystallinity → plateau ⑦ nearly flat in T until Tm.

(5) Below Tg the whole sample is already glassy → crystallinity has little effect.



① glassy ② transition ③ rubbery plateau ④ rubbery flow ⑤ viscous flow ⑥ crosslinked ⑦ semicrystalline

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Q32 *How many Maxwell models in parallel at least can you apply to describe a typical Master curve (Relaxation Modulus vs. Temperature)? How to match the curve?*

At least two Maxwell elements in parallel ($\tau_2 \gg \tau_1$). One element has one τ and can drop only once; the master curve drops twice (glass → rubber → flow).

$E(t) = E_1 e^{-t/\tau_1} + E_2 e^{-t/\tau_2}$, $\tau_i = \eta_i/E_i$

① $t \ll \tau_1$: $Eg = E_1 + E_2$.

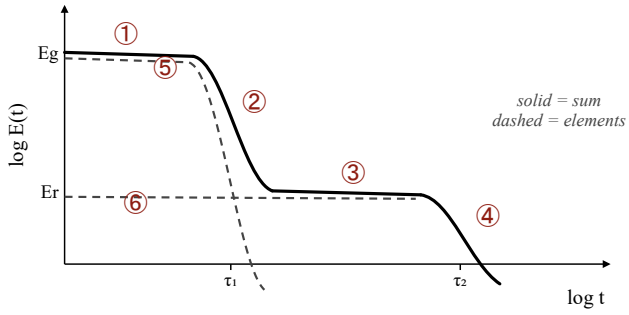
② $t \approx \tau_1$: fast element relaxes, drop of E_1.

③ $\tau_1 \ll t \ll \tau_2$: plateau $Er = E_2$.

④ $t \approx \tau_2$: falls to zero (terminal flow).

⑤⑥ (dashed) the two single elements.

Matching: step heights give E_1, E_2; drop positions give τ_1, τ_2.



Q40 *The viscosity of polystyrene is 10^{12} Pa·s at 373 K. Determine the viscosity when the temperature increases to 473 K (WLF eq., $a_1 = 17.44$, $a_2 = 51.6$)*

$\eta(473\ K) \approx 3.1$ Pa·s.

WLF with Tg = 373 K as reference, ΔT = 100 K:

$\log a_T = -17.44 \times 100 / (51.6 + 100) = -11.5 \rightarrow a_T = 3.1\times10^{-12}$

$\eta = a_T \times \eta(Tg) = 3.1\times10^{-12} \times 10^{12} \approx 3.1$ Pa·s

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8. Melt rheology

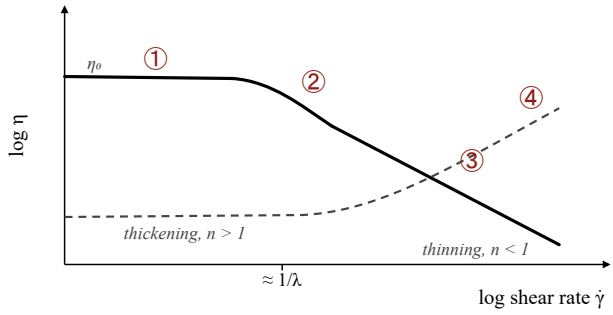
Q26 *What is shear thinning? What is shear thickening? Why a polymer behaves a shear thinning rather than shear thickening.*

Power law: $\eta = m\cdot\dot{\gamma}^{n-1}$ (n = 1: Newtonian).

Shear thinning: $n < 1$, η decreases as $\dot{\gamma}$ increases (solid line ③).

Shear thickening: $n > 1$, η increases as $\dot{\gamma}$ increases (dashed ④, e.g. concentrated suspensions).

Why polymers thin: at high $\dot{\gamma}$ the chains cannot relax, so they uncoil, align in the flow direction and disentangle → less flow resistance. There is no mechanism (like particle jamming) that would make them thicken.

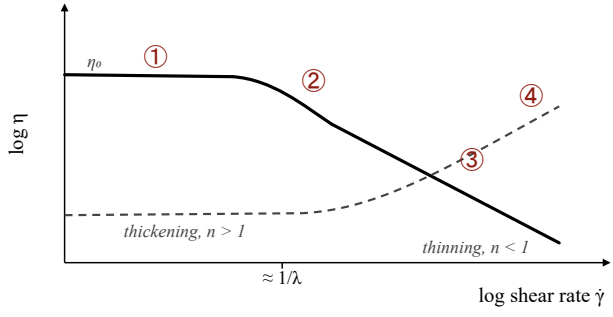


Q23 *Why does a polymer melt usually exhibit a shear thinning behavior?*

① Low $\dot{\gamma}$ ($\dot{\gamma} < 1/\lambda$): chains relax in time → Newtonian plateau η_0.

② $\dot{\gamma} \approx 1/\lambda$: the curve bends down.

③ High $\dot{\gamma}$: chains cannot relax → uncoil, align and disentangle → $\eta = m\cdot\dot{\gamma}^{n-1}$, $n < 1$ (shear thinning).



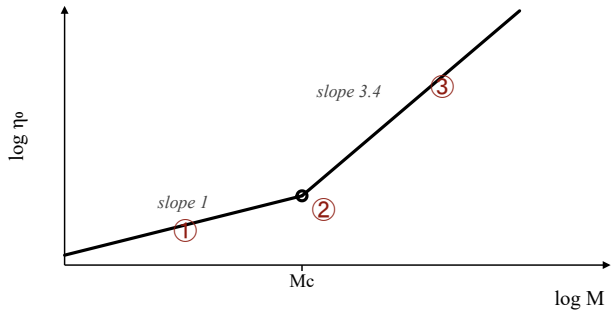
Q28 *What is the relationship between the molecular weight and the melt viscosity?*

① M < Mc: no entanglement, $\eta_0 \propto M$ (slope 1).

② Break at Mc ≈ 2–3 Me.

③ M > Mc: entanglement, reptation → $\eta_0 \propto M^{3.4}$ (slope 3.4).

Doubling M → $\eta_0 \times 2^{3.4} \approx 10.6$.



Q29 *What will the melt viscosity be affected by (1) temperature (2) pressure (3) shear rate (4) molecular weight (5) plasticizer wt%*

η decreases with (1) T↑, (3) $\dot{\gamma}$↑, (5) plasticizer↑; η increases with (2) pressure↑, (4) M↑.

(1) Higher T → more free volume → η↓ (WLF / Doolittle $\eta = A \exp(B/Vf)$).

(2) Higher pressure → less free volume → η↑.

(3) Higher $\dot{\gamma}$ → chains orient and disentangle → shear thinning.

(4) Higher M → more friction and entanglement: $\eta_0 \propto M$ (< Mc), $\propto M^{3.4}$ (> Mc).

(5) Plasticizer → more free volume, lower Tg → η↓.

Q30 *Please explain the Cox-Merz rule.*

Cox–Merz rule: the steady-shear viscosity equals the magnitude of the complex viscosity at the same numerical value of shear rate and angular frequency:

$$\eta(\dot{\gamma}) \approx |\eta^*(\omega)| \text{ at } \dot{\gamma} = \omega, \quad |\eta^*| = \sqrt{(G'^2 + G''^2)}/\omega$$

So a small-amplitude frequency sweep (G′, G″) can predict the steady flow curve (two different methods give nearly the same viscosity). It is empirical: works for homogeneous linear melts; may fail for filled systems, gels and block copolymers.

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9. Fracture mechanics

Q41 *Considering an aluminum plate has Kc of 44 MPa $m^{1/2}$, thickness 20 mm and width 100 mm, with 40 mm crack. What is the critical limit for the plate of load (kN)? (the geometry factor Y = 2.1)*

Fc ≈ 118 kN.

Fracture when K = Kc, with $K = Y\sigma\sqrt{(\pi a)}$, $\sigma = F/(WB)$:

$Fc = Kc \cdot W \cdot B / (Y\sqrt{(\pi a)}) = (44\times10^6 \times 0.100 \times 0.020) / (2.1 \times \sqrt{(\pi \times 0.040)})$

$= 88\,000 / (2.1 \times 0.3545) \approx 1.18\times10^5$ N ≈ 118 kN
